

PENGWIN 2026 Task 1 -- V0.2 Pipeline-Test Submission

Algorithm Description · 2026-05-30

0. Submission intent

This is a **V0.2 pipeline-test** submission for the PENGWIN 2026 Task 1 preliminary phase. The objective is to validate the end-to-end Grand Challenge contract (Docker container build, model tarball, /input MHA read, /output MHA write under `--network none`) with the current best in-house two-stage pipeline. It is **not** a competitive scientific submission and should not be interpreted as our final leaderboard entry. The femur anatomy (label range 151-200) is **intentionally not modeled** in V0.2; femur fragments are emitted as background (0). Femur modeling is planned for V1.

1. Method overview

V0.2 is a **two-stage anatomy-conditioned pipeline**. Stage 1 is a whole-CT anatomy classifier that produces a 4-class softmax (background / Sacrum / LeftHipbone / RightHipbone). Stage 2 is a per-anatomy ABBC nnU-Net that takes a **3-channel ROI** -- the bone-LUT CT crop, the Stage 1 anatomy probability crop, and a signed distance map derived from that probability -- and produces a 4-class ABBC softmax (background / border / boundary / core). Per-instance fragments are recovered by a core-seed watershed decoder and pasted back into the full label volume.

Components:

- **Stage 1 -- anatomy classifier:** nnU-Net ResEnc-L trained on Dataset532_PelvicAnatomyV2. Input is the full CT in LPS orientation, normalized with the bone HU window $[-1000, 2000]$ clip + bone-LUT. Output is a 4-class softmax over background / Sacrum / LeftHipbone / RightHipbone.
- **Per-anatomy ROI extraction:** for each of {Sacrum, LeftHipbone, RightHipbone}, take the bounding box of the Stage 1 prob ≥ 0.5 mask, pad by 24 voxels on each axis, and build a **3-channel ROI**: 1. bone-LUT normalized CT crop, 2. Stage 1 anatomy probability crop (the channel for this anatomy), 3. signed distance to the prob ≥ 0.5 surface, clipped to ± 40 mm.
- **Stage 2 -- ABBC fragment model:** V291 ABBC nnU-Net ResEnc-L (nnUNetResEncUNetLPlans, 3d_fullres), 3-channel input, 4-class ABBC softmax (background / border / boundary / core).
- **Instance decoder:** V288-style core-seed watershed lifts the 4-class softmax to per-instance integer fragments, followed by $\sim 1 \text{ cm}^3$ connected-component prune and an anatomy-size-ratio merge (`sr_keep = 0.05`).
- **Label assembly:** per-anatomy fragments are remapped into the PENGWIN label ranges (Sacrum 1-50, LeftHipbone 51-100, RightHipbone 101-150) and pasted into the full-volume label map. The label map is then reoriented back from LPS to the original input frame.

Femur (151-200) is **intentionally unmodeled** in V0.2 and is written as background; a femur stage is planned for V1.

1.5 V0.3 robustness layers

V0.3.1 wraps the V0.2 two-stage pipeline in a **4-layer robustness stack** designed to keep the container alive and on-budget across out-of-distribution Grand Challenge inputs. V0.2 had a known GC failure mode: on small-FOV inputs the Ds532 Stage 1 anatomy classifier fired out-of-distribution and produced large spurious anatomy masks, which caused the per-anatomy ROI to balloon, blow past the 10-minute GC time budget, and time out the case. V0.3.1 fixes this by adding distribution-independent fallbacks, mutually-exclusive anatomy assignment, post-pad bbox sanity, and an explicit time budget.

- **L1 -- bone-skeleton anatomy decomposition** ($HU > 200$ connected components). A distribution-independent fallback derived directly from the CT (bone-threshold + 3D connected components) that produces a coarse Sacrum / LeftHipbone / RightHipbone partition without depending on the Ds532 classifier. L1 is always available as a fallback when L2 misfires or is suppressed by L3.
- **L2 -- Ds532 argmax masks** (mutually-exclusive anatomy assignment). Replaces the V0.2 per-channel prob ≥ 0.5 threshold (which could double-assign voxels to overlapping anatomies and produce inflated ROIs) with a single argmax over the 4-class Stage 1 softmax, yielding mutually-exclusive Sacrum / LeftHipbone / RightHipbone masks.

- **L3 -- bbox-fraction sanity** (post-pad bbox >50% triggers pad-shrink: 24 -> 12 -> 6 -> 0 vox). After padding the L2 argmax bounding box by 24 voxels, if the resulting box covers more than 50% of the full CT volume the pad is shrunk through the cascade 24 -> 12 -> 6 -> 0 voxels until the post-pad bbox falls under the 50% threshold. This catches Ds532 OOD misfires before they can enter Stage 2.
- **L4 -- time-budget enforcement** (8 min soft, 9 min hard). A wall-clock budget tracked across the per-anatomy loop: at 8 minutes the pipeline switches to a soft-degraded path (skips remaining optional refinement), at 9 minutes it hard-stops and emits the best label map assembled so far rather than timing out the GC job at 10 minutes.

V0.3.1 directly addresses V0.2's GC failure mode -- Ds532 OOD misfire on small-FOV inputs causing a 10-minute timeout -- through the L2 argmax fix (no double-assignment), the L3 bbox-fraction sanity check (shrinks the pad before the ROI blows up), the L1 bone-skeleton fallback (usable when L2/L3 reject the Ds532 output), and the L4 explicit time budget (graceful degradation instead of timeout).

1.5b V0.3.3 morphology cleanup + bone fallback after bbox sanity fail

V0.3.3 patches the L2 + L3 path to harden the Ds532 argmax masks against a specific Ds532 failure mode where the dominant CC passes post-pad bbox sanity but is contaminated by sparse outlier CCs (thin streaks / isolated voxels) that survive the largest-CC keep heuristic via voxel-adjacent bridges. Two changes:

- **Ds532 mask morphology cleanup.** After argmax, each anatomy mask is passed through `binary_opening` (one iteration, 2x2x2 structure) and then a small-CC drop step (CCs with fewer than `MIN_DS532_CC_VOXELS = 500` voxels are removed). This is a benign cleanup on well-behaved cases (the dominant CC is unchanged) and a meaningful denoising on cases where Ds532 emits sparse outlier streaks.
- **Bone-skeleton fallback after post-pad bbox-sanity fail.** If the L3 pad-shrink cascade (24 -> 12 -> 6 -> 0 voxels) exhausts all four pad values and the post-pad bbox is still above the 50% threshold, V0.3.3 attempts the L1 bone-skeleton mask (largest CC only) before emitting zero for that anatomy. If the bone-skeleton bbox passes sanity, it is used as the Stage 2 ROI; otherwise the anatomy is emitted as zero (V0.3.1 behavior).

A/B (V0.3.3 vs V0.3.1) on a 3-case small-FOV pelvic slice (physical-x 248-258 mm; cases 162, 163, 189) shows **zero cohort delta on all metrics** (Fracture Dice, Instance F1, HD95 all 0.000). The morphology cleanup fires on case 162 (5 sparse CCs dropped on Sacrum, 5 on RightHip, 1 on LeftHip; 600-1000 voxels removed per anatomy) but the dominant CC is unchanged, so the painted ROI and downstream metrics are byte-identical to V0.3.1. The bone-skeleton fallback path does not fire on this slice because the Ds532 sanity check already passes at the L3 stage. V0.3.3 is therefore a **safety-net change**: benign on the tested cohort (no regression), but provides defense-in-depth against a known Ds532 sparse-outlier failure mode that does not appear in the held-out probe set.

1.5c V0.3.4 bone-skeleton-aware pelvic/femur routing fix

V0.3.4 patches the top-level pelvic-vs-femur anatomy routing (`classify_pelvic_femur`) that runs **before** the L1-L4 robustness stack. The V0.3.1 routing rule was purely spacing-and-physical-FOV based, which misrouted small-FOV pelvic cases with `spacing_z = 0.80 mm` to the femur branch -- and since femur is intentionally unmodeled in V0.x, the container wrote a zero-valued label map and scored zero on the case. Case 189 (small-FOV pelvic, `spacing_z = 0.80`) is the canonical reproducer of this misroute.

- **Bone-skeleton-aware router (`classify_pelvic_femur_v2`).** Before falling back to the spacing rule, the router runs the L1 bone-skeleton anatomy decomposition (`HU > 200` connected components, sized by `BONE_MIN_COMPONENT_VOXELS`) on the LPS-canonicalized + HU-clipped CT and counts how many of {Sacrum, LeftHip, RightHip} are recovered with at least `BONE_MIN_COMPONENT_VOXELS` voxels. If **two or more** valid pelvic anatomies are detected, the case is force-routed to pelvic regardless of spacing/FOV. Otherwise the original V0.3.1 spacing rule is used as fallback.
- **Mask reuse into L1.** When the router force-routes to pelvic, the bone-skeleton masks it computed are threaded through into `run_per_anatomy_pelvic` as `prerouted_bone_masks`, so the L1 step inside the per-anatomy pipeline reuses the already-computed masks instead of recomputing them. This keeps the V0.3.4 path **bit-identical to V0.3.3** on cases that the V0.3.1 spacing router was already routing to pelvic correctly.

A/B (V0.3.4 vs V0.3.3) on the same 3-case small-FOV pelvic slice (cases 162, 163, 189): **one routing change** -- case 189 is rescued from the femur branch (V0.3.3 wrote zero output) to the pelvic branch (V0.3.4 writes ~11.2M nonzero voxels through the full V291 ABBC pipeline). Cases 162 and 163 route to pelvic in both versions and are byte-identical. The

routing fix itself is verified to work end-to-end on case 189.

Cohort delta (V0.3.4 vs V0.3.3, n=3, cases 162/163/189): Fracture Dice 0.000 -> 0.000 (delta 0.000), Instance F1 0.000 -> 0.000 (delta 0.000), HD95 undefined in both versions. The headline-metric delta is **zero** even though case 189 is now routed correctly and the V291 decoder runs on it. The reason is a model/decoder accuracy issue, not a routing issue: on this small-FOV slice the V291 ABBC decoder paints fragments that are substantially larger than the GT fragments (IoU ~0.06-0.08, below the `iou_match_threshold = 0.10` used by the per-anatomy argmax matching in the official-aligned eval), so no GT fragment acquires a matched prediction and the per-case Fracture Dice and Instance F1 collapse to 0. V0.3.4 is therefore a **routing-correctness fix**: it eliminates a known failure mode (small-FOV pelvic misrouted to zero output), is verified to reroute case 189 to the pelvic branch and run the full V291 pipeline on it, and does not regress any case in the held-out or probe cohorts. The remaining decoder-accuracy gap on this small-FOV slice is a Stage 2 model issue and is out of scope for the V0.3 robustness stack; it is tracked for V1 fulltrain.

V0.3.4 is the configuration actually shipped in this submission's container. The headline V0.3.1 held-out numbers in Section 5b remain the leaderboard projection for V0.3.4, because the V0.3.4 routing change does not fire on any of the V0.3.1 held-out cases (001, 002, 004, 005, 006) -- they were all already routed to pelvic by the V0.3.1 spacing rule -- so V0.3.4 is **bit-identical to V0.3.3 (and to V0.3.1 on this cohort)** on the held-out cohort. V0.3.4 is a strictly additive safety-net change for the small-FOV pelvic misroute failure mode.

2. Training

Item	Value
Stage 1 dataset	Dataset532_PelvicAnatomyV2 (background / Sacrum / LeftHipbone / RightHipbone)
Stage 2 dataset	Dataset537_PelvicBICMFragmentV5 (Sacrum / LeftHipbone / RightHipbone; no Femur)
Stage 2 trainer	PenguinTrainerBoundaryFragmentSidecarCoreRecallDenseCandidateCore025StrongPeakNoContactABBCV291
Plans	nnUNetResEncUNetLPlans
Config	3d_fullres
Stage 2 input channels	3 (bone-LUT CT, Ds532 anatomy prob, signed distance)
Fold	fold_0 only
Boundary class weight (bw)	10
Epochs	50
Iterations / epoch	100 train / 20 val
Train ROIs	12 stratified (sub-fulltrain probe)
Val ROIs	6 stratified
Best EMA val score	0.188 (best checkpoint at epoch 34 spike)
Stage 2 checkpoint	checkpoint_best.pth (~819 MB)

The Stage 2 model was trained as a **probe** for the V291 ABBC head design, not as a full-data training run. A Phase 6 fulltrain on all 174 cases is planned but is not part of this submission.

3. Inference

3.1 Pipeline

1. **Read** input MHA from `/input/images/peripelvic-fracture-ct/`.
2. **LPS canonicalization** -- reorient the input volume into LPS and record the original orientation transform.
3. **Bone HU clip + LUT normalize** -- clip CT to `[-1000, 2000]` and apply the bone-window LUT (used by both stages).

4. **Stage 1 (Ds532 anatomy classifier)** -- nnU-Net ResEnc-L predict on the full CT, producing a 4-class softmax (background / Sacrum / LeftHipbone / RightHipbone).
5. **Per-anatomy ROI loop** for each of {Sacrum, LeftHipbone, RightHipbone}: a. Compute the bounding box of prob ≥ 0.5 for this anatomy and pad by 24 voxels on each axis. b. Build a 3-channel ROI input:
 - channel 0: bone-LUT CT crop,
 - channel 1: Stage 1 anatomy probability crop,
 - channel 2: signed distance to the prob ≥ 0.5 surface in mm, clipped to ± 40 mm. c. **Stage 2 (V291 ABBC) predict** -- nnU-Net ResEnc-L on the 3-channel ROI, producing a 4-class ABBC softmax (background / border / boundary / core). d. **Core-seed watershed** (V288-style decoder) lifts the ABBC softmax to per-instance integer fragments. e. **Connected-component prune** at $\sim 1 \text{ cm}^3$ removes speckle. f. **Anatomy-size-ratio merge** with $\text{sr_keep} = 0.05$ (fragments below 5% of the largest fragment within an anatomy are merged into the nearest large fragment). g. **Re-label** the fragments into the PENGWIN range for this anatomy: Sacrum 1-50, LeftHipbone 51-100, RightHipbone 101-150. h. Paste the relabeled ROI into the full-volume LPS label map.
6. **Femur (151-200)** is not modeled in V0.2 and remains 0.
7. **Reorient** the label volume from LPS back to the original input frame.
8. **Write** the integer 3D label map MHA to `/output/images/peripelvic-fracture-ct-segmentation/`, preserving the input spacing, dimensions and orientation.

The runtime entrypoint is `inference/inference.py`, packaged at `/opt/app/` in the container. Stage 1 (Ds532) and Stage 2 (V291) checkpoints, `plans.json`, and `dataset_fingerprint.json` are shipped via the model tarball, extracted by Grand Challenge under `/opt/ml/model/`.

4. Known limitations

- **Femur not modeled.** Femur fragments (label range 151-200) are written as background in V0.2. Any case whose ground-truth contains labels in [151, 200] will score zero on those instances. Femur is planned for V1.
- **Sub-fulltrain probe.** Only 12 of 174 cases were used to train the Stage 2 model; the EMA val score (0.188) and the hard-trio metrics (below) are well below the leaderboard target. Hard-trio Fracture Dice 0.812 is below the ~ 0.90 leaderboard expectation.
- **Single fold, single seed.** No ensembling, no TTA beyond nnU-Net defaults.
- **Phase 6 fulltrain planned.** The next submission will train on all 174 cases with both pelvic and femur anatomies and a full 1000-epoch schedule.

5. V0.2 held-out cohort metrics

Metrics are computed by our internal CLI `task1-v288-eval-v2` (`compute_task1_official_aligned_v2_metrics`), which is aligned to the PENGWIN 2026 official scoring spec.

The honest held-out cohort is the **five cases that are out-of-fold0 for the V0.2 Stage 2 model**: 001, 002, 004, 005, 006. These are the only cases in this section that the model has never seen during training; they are the relevant signal for what the preliminary leaderboard will look like. Case 003 was inside the fold0 train set and is shown alongside as a sanity check, not as a leaderboard estimate.

Cohort	n	Fracture Dice	Local Dice (20mm)	HD95 (mm)	ASSD (mm)	Instance F1	Instance Recall	Instance Precision
Held-out (out-of-fold0)	5 (001/002/004/005/006)	0.612	0.616	26.7	5.7	0.683	0.785	0.708
Train leak (in fold0 train)	1 (003)	0.827	0.828	3.8	1.5	0.808	0.952	0.778

The held-out row is the configuration actually shipped in the V0.2 container (**bw=10, sr_keep=0.05**, V291 ABBC Stage 2 + Ds532 Stage 1). The 003 train-leak row is shown only for comparison and must not be read as a held-out estimate.

GC preliminary score expectation: Fracture Dice ~0.55-0.65, Instance F1 ~0.60-0.70 (femur cases 151-200 will dilute the cohort because V0.2 emits zeros for that range).

Footnote -- *training-set fit only, not held-out*: an earlier Phase 5.5 v2 hard-trio (cases 003, 011, 017, all in fold0 train per splits_final.json) reported Fracture Dice **0.812**, Local Dice **0.819**, Instance F1 **0.890**, HD95 **11.79 mm**, ASSD **2.59 mm**, Merge Err **2**, Split Err **21**, Topology Cons. **0.444** for the **bw 10 / sr 0.05 (V0.2)** configuration. These three cases are inside the fold0 training set and therefore are **not a held-out estimate**; they are retained here only to document the configuration sweep and should be ignored when projecting leaderboard performance. Alternate ablation configurations (bw 2.5 / sr 0.00, bw 2.5 / sr 0.05, bw 10 / sr 0.00) were run internally but are not reported here; only bw=10 / sr_keep=0.05 is shipped in the container.

5b. V0.3.1 held-out cohort metrics (this submission)

This is the **headline** held-out result for the V0.3.1 container that is actually shipped. The held-out cohort is identical to the V0.2 cohort -- the **five out-of-fold0 cases** 001, 002, 004, 005, 006 -- so the V0.3.1 row below is a like-for-like replacement of the V0.2 held-out row in Section 5; the V0.2 table is retained above for diff/context only.

Cohort	n	Fracture Dice	Local Dice (20mm)	HD95 (mm)	ASSD (mm)	Instance F1	Instance Recall	Instance Precision
V0.3.1 held-out (out-of-fold0)	5 (001/002/004/005/006)	0.621	0.616	20.6	4.96	0.805	0.774	0.933
V0.2 held-out (out-of-fold0)	5 (001/002/004/005/006)	0.612	0.616	26.7	5.72	0.683	0.785	0.708
Delta (V0.3.1 - V0.2)	--	+0.010	0.000	-6.1	-0.76	+0.123	-0.011	+0.225
Train leak (in fold0 train, 003)	1 (003)	0.827	--	3.8	--	0.974	--	--

The V0.3.1 row is what we expect the preliminary leaderboard to look like for this submission. The Train-leak row (case 003) is shown only as a sanity check that the model has not regressed on training data; it is **inside the fold0 train set** and must not be read as a held-out estimate.

Headline movements vs V0.2:

- **Fracture Dice 0.612 -> 0.621** (+0.010): small overall-overlap improvement.
- **Local Dice (20 mm) 0.616 -> 0.616** (0.000): unchanged, as expected -- L1-L4 robustness layers do not change the local 20 mm overlap structure on cases where V0.2 already produced a reasonable ROI.
- **HD95 26.7 mm -> 20.6 mm** (-6.1 mm) and **ASSD 5.72 mm -> 4.96 mm** (-0.76 mm): boundary-distance metrics improve substantially, consistent with L3 bbox-fraction sanity removing spurious far-field fragments.
- **Instance F1 0.683 -> 0.805** (+0.123) driven almost entirely by **Instance Precision 0.708 -> 0.933** (+0.225) while **Instance Recall 0.785 -> 0.774** (-0.011) stays roughly flat. This is the signature of the L2 argmax + L3 bbox-fraction stack: false positives from Ds532 OOD blow-ups are eliminated, recall is essentially preserved.

Per-case (V0.3.1 held-out):

Case	Fracture Dice	Instance F1	HD95 (mm)	Notes
001	0.620	0.667	15.4	--
002	0.688	0.889	8.0	--
004	0.483	0.822	29.5	hardest case

Case	Fracture Dice	Instance F1	HD95 (mm)	Notes
005	0.750	1.000	38.6	--
006	0.566	0.650	11.7	--

Case 004 is the hardest of the held-out cohort (lowest Fracture Dice at 0.483, highest HD95 at 29.5 mm), although it still attains a respectable Instance F1 of 0.822. Case 005 attains perfect Instance F1 (1.000) but with a high HD95 (38.6 mm), indicating that fragment identities are correctly recovered while the boundary location of one or more fragments is locally off.

Train-leak (case 003, **training-set fit only, not held-out**): Fracture Dice **0.827**, Instance F1 **0.974**, HD95 **3.8 mm**. This case was inside the fold0 training set and is reported only as a sanity check that the V0.3.1 robustness layers (L1-L4) do not regress training-set fit. It must **not** be read as a held-out estimate or as a leaderboard projection.

V0.3.1 is the configuration actually shipped in this submission's container: V0.2 two-stage pipeline (Ds532 Stage 1 + V291 ABBC Stage 2, **bw=10, sr_keep=0.05**) wrapped in the L1-L4 robustness stack described in Section 1.5. The V0.3.1 numbers above supersede the V0.2 numbers in Section 5 as the headline held-out result for this submission; the V0.2 table is retained above only to make the diff explicit.

6. Pointers

- Stage 1 anatomy dataset: Dataset532_PelvicAnatomyV2 (background / Sacrum / LeftHipbone / RightHipbone).
- Reference baseline: YzzLiu/PENGWIN2026_Task1_AutoSeg_Baseline (preprocessing / training / inference scaffolding; no evaluate.py).
- Internal eval CLI used to produce the metrics in Section 5: task1-v288-eval-v2 (compute_task1_official_aligned_v2_metrics).
- Internal predict CLI mirrored by the container entrypoint: task1-v288-predict (code_task1/eval.py, --cases / --out-root / --checkpoint / --fold / --trainer / --plans / --config / --preprocessed-plans-dir).